

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

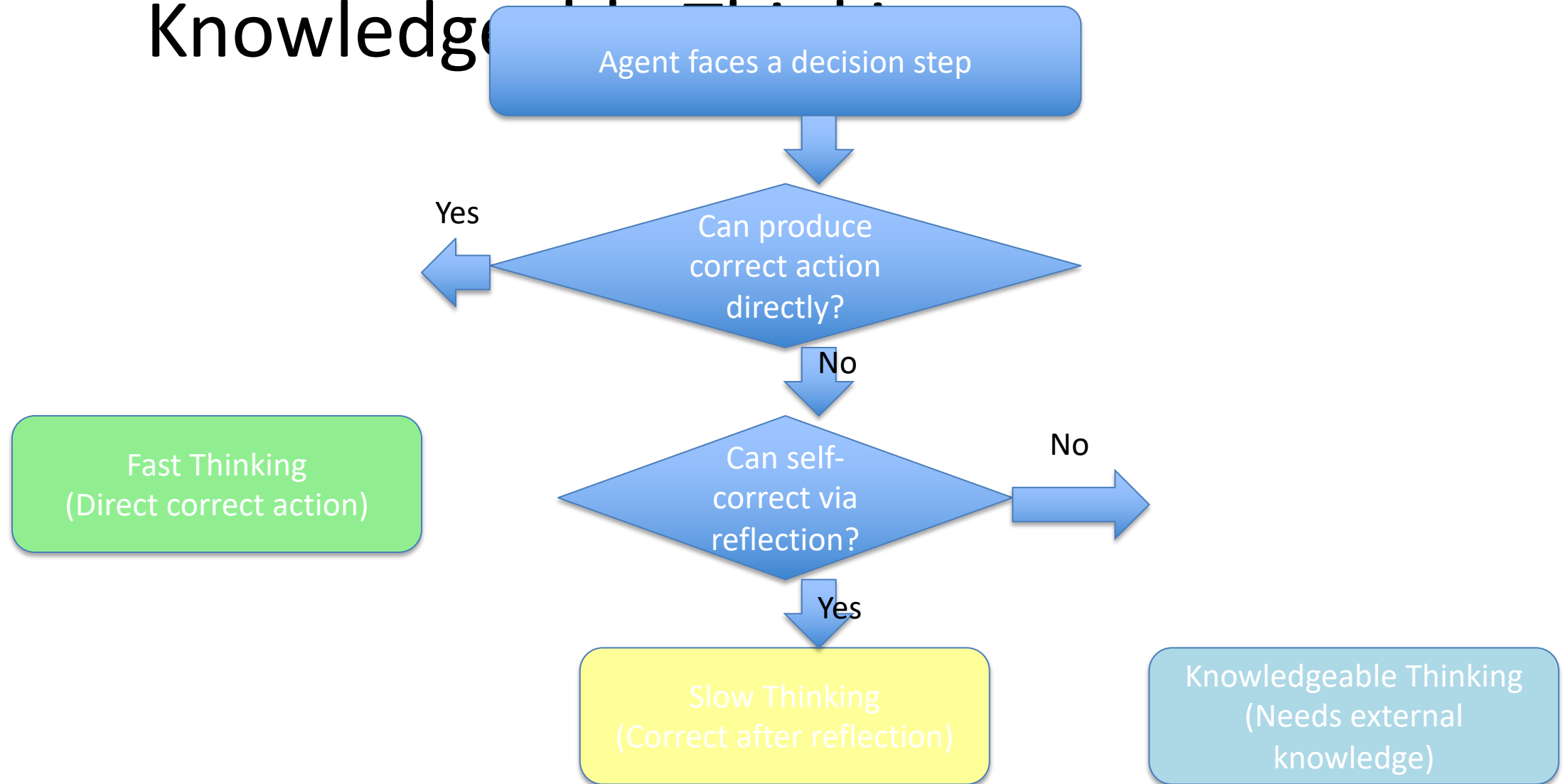
Teaching LLM Agents When to Think,
Reflect, or Seek Knowledge

arxiv.org/abs/2504.03553

Irrigation' — Indiscriminate Knowledge Injection Hurts

- **The Problem: Indiscriminate Knowledge Injection in Agent Planning**
 - Existing approaches use a 'flood irrigation' strategy, blindly injecting resources regardless of situational needs.
- **Three Existing Paradigms & Limitations:**
 - 1. Direct Trajectory Imitation (e.g., ReAct): Mimics patterns without deep understanding, leading to fragile planning.
 - 2. Trial-and-Error Refinement (e.g., Reflexion): Unnecessarily high inference costs for simple

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training

Step 1 — Self-Awareness Data Construction → Self-Aware Inference

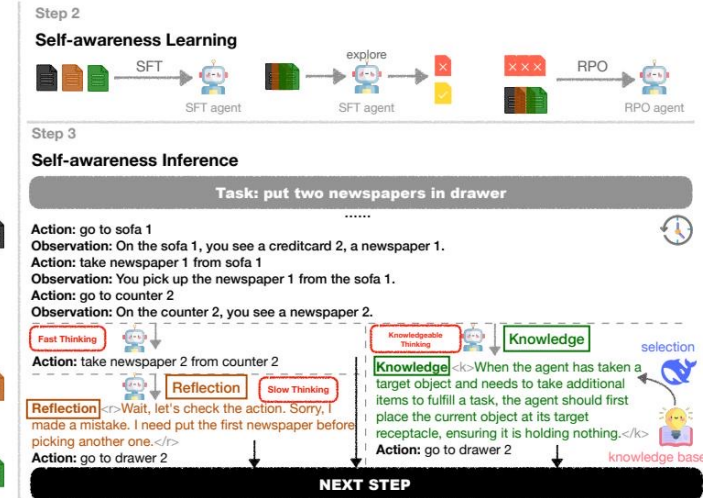
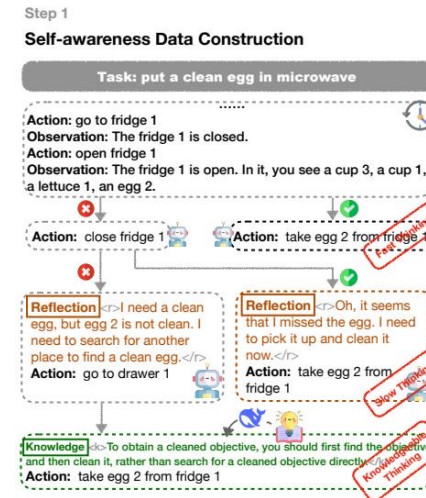
Agent self-explores environments. A heuristic criterion classifies the situation (fast/slow/knowledgeable) based on prediction vs gold action. Special tokens mark each type.

Step 2 — Two-Stage Training

1. Supervised Fine-Tuning (SFT) teaches initial self-awareness patterns.
2. Reward-based Policy Optimization (RPO) refines token generation.

Step 3 — Self-Aware Inference

Agent generates special tokens that trigger direct action, reflection, or retrieval from an external knowledge base.



KnowSelf Correctly Heats a Mug and Places a Saltshaker Where

Baselines Fail

Example A: 'Put a hot mug in cabinet'

- OpenAI-O1: Incorrectly tries to take an egg from the microwave after opening it.
- KnowSelf (Llama-8B): Uses a Reflection token to realize it should heat the mug using the open microwave.

Example B: 'Put a saltshaker in drawer'

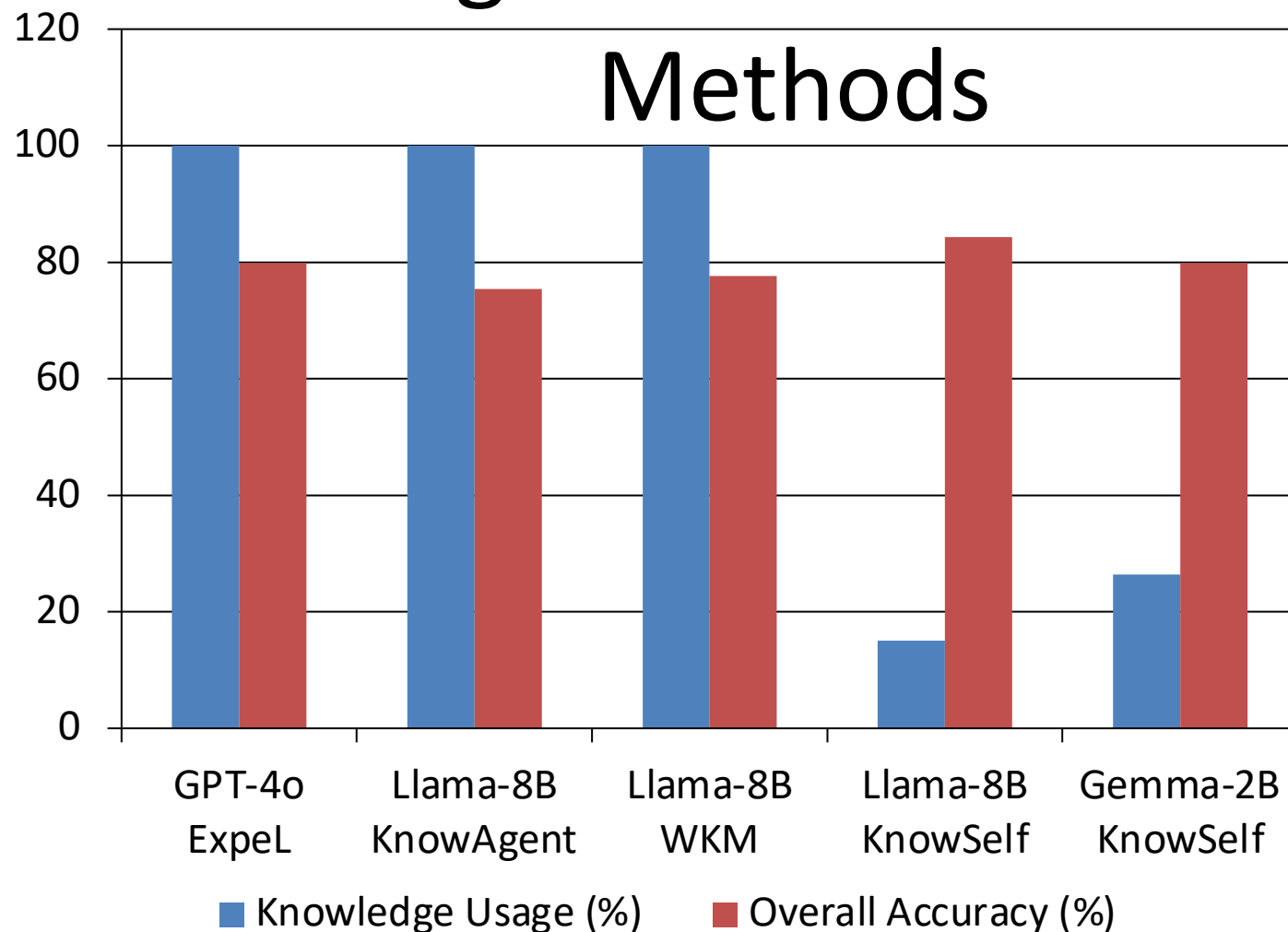
- DeepSeek-R1: Reflects that the drawer is open and tries to close it first (fails).
- KnowSelf (Llama-8B): Applies a Knowledge token noting it should place the object without removing unrelated items.

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</> Action: put saltshaker 1 in/on drawer 1

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge



Key Efficiency Insights:

- SOTA Performance: KnowSelf achieves the highest overall scores while using only a fraction of external knowledge (~15-26%).
- The 'Flood Irrigation' Trap: Methods with 100% knowledge usage (ExpeL, KnowAgent, WKM) underperform KnowSelf, showing indiscriminate knowledge injection is actively detrimental to reasoning flow.
- Efficient Scaling: Gemma-2B matches the performance of the massive GPT-4o ExpeL model.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- 1. Indiscriminate knowledge injection is wasteful and suboptimal.
- 2. Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- 3. KnowSelf with Llama-8B outperforms strong baselines including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).
- 4. Only 15-26% knowledge usage needed versus 100% for competing methods.